

# DESIGN BREAKDOWN - LITERAL VISUAL EXPLANATION

## What You're Describing (No Code, Just Design)

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### THE CORE PROBLEM YOU'RE SOLVING

#### Right now:

- Temperature data exists for specific geographic areas
- Map shows orange/yellow/red PATCHES floating on the map
- These patches DON'T connect to any geographic boundary
- User sees: "Why is there a red dot here? What area does it cover?"
- **The patches look random because they have no geographic anchor**

#### What you want:

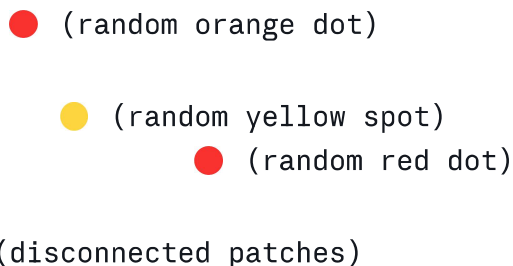
- Temperature data is tied to SPECIFIC REGIONS (LGAs, clusters)
  - Map shows COLORED REGIONS with clear boundaries
  - Each boundary = a real geographic area
  - User sees: "This region is red = hot. This region is yellow = cool."
  - **The colors make sense because they're inside defined boundaries**
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### THE DESIGN CONCEPT (Visual Relationship)

#### BEFORE (What she built):

##### MAP VIEW:

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- 
- (random orange dot)
  - (random yellow spot)
  - (random red dot)
  - (disconnected patches)
- 

##### PROBLEM:

- Patches float on the map

- No connection to any region
- No clear boundaries
- Looks like random data points

## AFTER (What you want):

MAP VIEW:

|                                                                        |                                                                        |
|------------------------------------------------------------------------|------------------------------------------------------------------------|
| <div><div>● RED</div><div>(HOT ZONE)</div><div>28-30°C</div></div>     | <div><div>● YELLOW</div><div>(COOL ZONE)</div><div>24-26°C</div></div> |
| <div><div>● ORANGE</div><div>(WARM ZONE)</div><div>26-28°C</div></div> | <div><div>● CREAM</div><div>(MODERATE)</div><div>27-29°C</div></div>   |

SOLUTION:

- Each region has clear boundaries
- Colors FILL the entire region
- Boundaries show what geographic area each color represents
- Looks like proper data mapping

## THE VISUAL RELATIONSHIP YOU'RE DESCRIBING

### CONCEPT 1: Regions as Containers

TEMPERATURE DATA → LIVES INSIDE → REGION BOUNDARY  
(Polygon/Shape)

Example:

"Katsina has 28°C temperature" → FILLS the entire Katsina LGA region with ORANGE color

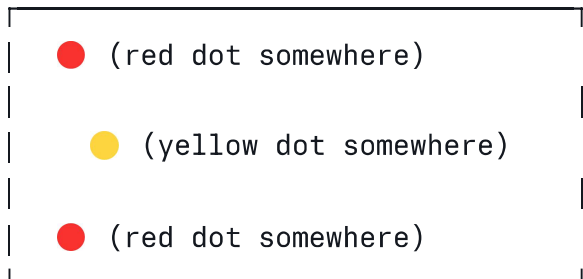
"Kano has 31°C temperature" → FILLS the entire Kano region with RED color

## What the user sees:

- Not a dot on Katsina
  - But the ENTIRE Katsina region is colored orange
  - The color extends to the region's edges
  - Clear visual relationship: Region = Color = Temperature
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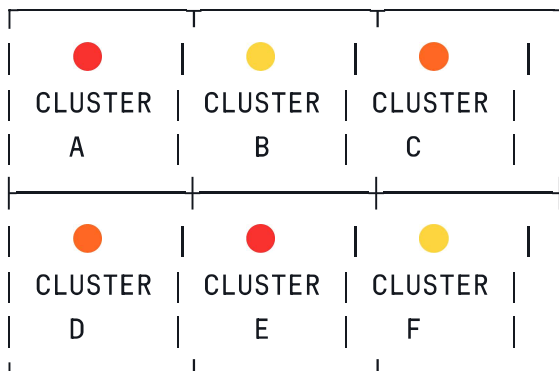
## CONCEPT 2: Boundaries Create Structure

WITHOUT BOUNDARIES:



Result: Looks random, confusing

WITH BOUNDARIES (Sub-clusters):



Result: Clear structure, easy to understand

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## THE SPECIFIC VISUAL TRANSFORMATION

### STEP 1: Start with LGA/Region Boundaries

Map of Nigeria showing LGA boundaries as BLACK LINES

Katsina LGA:

(empty region)

Kano LGA:

(empty region)

Etc. for all 6 states

**What you're showing:** The geographic grid (bones of the map)

## STEP 2: Add Temperature Data as Color

Now FILL each region with a color based on its temperature:

Katsina LGA:

● ● ● ● ●  
● ● ● ● ●  
● ● ● ● ●

← FILLED with ORANGE  
(28°C = Warm)

Kano LGA:

● ● ● ● ●  
● ● ● ● ●  
● ● ● ● ●

← FILLED with RED  
(31°C = Hot)

Etc.

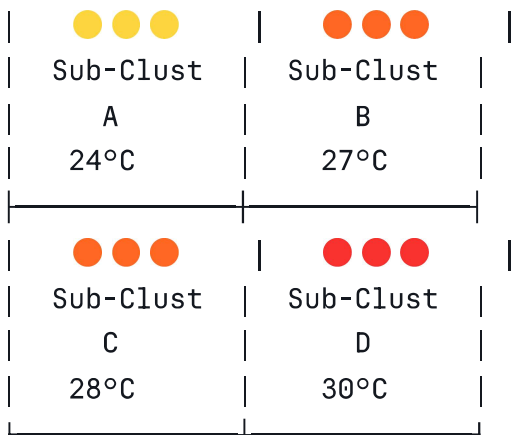
**What you're showing:** Each region's entire area is one color (its temperature)

## STEP 3: Show Color Gradations

ZONED IN VIEW (showing subdivisions/sub-clusters):

Katsina region divided into 4 sub-clusters:

| | | |



Each sub-cluster has its OWN SHADE:

- Light yellow = 24°C (coolest)
- Cream = 25°C
- Light orange = 26°C
- Orange = 27°C
- Dark orange = 28°C
- Red = 29°C
- Dark red = 30°C
- Crimson = 31°C

**What you're showing:** Each smaller region has a shade matching its specific temperature

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## THE COLOR SPECTRUM YOU WANT

SLIDER AT BOTTOM (Clean UI):

24°C ————— 35°C+

|                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
|  |  |  |  |  |
| Light                                                                               | Cream                                                                               | Orange                                                                              | Red                                                                                 | Dark                                                                                |
| Yellow                                                                              |                                                                                     |                                                                                     |                                                                                     | Red                                                                                 |

BUT BACKEND IS COLORING WITH 100+ SHADES:

24.0°C → #FFFACD (very light yellow)  
24.5°C → #FFF8DC (slightly less light)  
25.0°C → #FFE4B5 (cream)  
25.5°C → #FFDAB9 (peach)  
26.0°C → #FFD700 (golden)  
26.5°C → #FFA500 (orange)  
27.0°C → #FF8C00 (dark orange)  
27.5°C → #FF7F50 (coral)

28.0°C → #FF6347 (tomato red)  
28.5°C → #FF4500 (dark red)  
29.0°C → #CD5C5C (red)  
...continuing...  
35°C+ → #8B0000 (very dark crimson)

USER SEES: Smooth gradient

BACKEND KNOWS: Precise color for each temperature value

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## THE RELATIONSHIP BETWEEN VISUALS AND REGIONS

### WHAT THE USER UNDERSTANDS:

| VISUAL ELEMENT      | → WHAT IT MEANS                           |
|---------------------|-------------------------------------------|
| Colored Region      | → A geographic area with that temperature |
| Light Yellow Region | → Cool area (24-26°C)                     |
| Orange Region       | → Warm area (27-29°C)                     |
| Red Region          | → Hot area (30-32°C)                      |
| Dark Red Region     | → Extreme heat (33°C+)                    |
| Region Boundary     | → Edge of this cluster/LGA                |
| Region Size         | → How large this rice cultivation area is |
| Color Intensity     | → How hot/cool compared to neighbors      |

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## THE ZOOM EXPERIENCE

### ZOOMED OUT (State Level):

View entire state as ONE blended color  
User sees: "Kano is mostly red = hot state overall"

#### STATE BOUNDARY

|                                                               |
|---------------------------------------------------------------|
| Red blended across entire area<br>(30°C average across state) |
|---------------------------------------------------------------|

View individual clusters with their specific shades

User sees: "This cluster is orange, that one is red, this one is yellow"

CLUSTER BOUNDARIES (30-40 per state)

|                                                                                   |                                                                                   |                                                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                                                                   |                                                                                   |                                                                                   |
|  |  |  |
| 28°C                                                                              | 31°C                                                                              | 25°C                                                                              |
| Cluster                                                                           | Cluster                                                                           | Cluster                                                                           |
| A                                                                                 | B                                                                                 | C                                                                                 |

Each cluster is a DISTINCT COLORED REGION

Not a dot, not a patch

But a full polygon with boundaries

## THE DESIGN PRINCIPLE

### Three Things Must Align:

1. GEOGRAPHIC BOUNDARY (polygon shape)

↓

2. TEMPERATURE VALUE (data point)

↓

3. COLOR SHADE (visual representation)

Example:

Cluster "Katsina\_North"

(geographic boundary = polygon)

↓

Temperature = 28°C

(data point)

↓

Color = Orange #FFA500

(visual)

RESULT:

The entire "Katsina\_North" polygon is FILLED with orange

User sees: Orange shape on map = "This cluster is 28°C"

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## THE VISUAL CLARITY YOU'RE CREATING

### Before (Random Patches):

- User sees: "Orange and red dots scattered on map"
- Question: "What region are these?"
- Answer: "Unclear"

### After (Regions with Boundaries):

- User sees: "Orange and red REGIONS with clear edges"
  - Question: "What region is this?"
  - Answer: "Katsina cluster A, 28°C" (clear boundary = clear region)
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## WHAT THE NEW DEVELOPER NEEDS TO UNDERSTAND

### Design Spec (No Code):

#### WHAT TO BUILD:

1. START WITH: Polygon boundaries for each cluster/LGA  
(These are the SHAPES on the map - the container)
2. GET: Temperature data for each cluster  
(This is the DATA - the measurement)
3. ASSIGN: A color shade to each temperature  
(This is the COLOR - the visual)
4. FILL: Each polygon boundary with its color  
(This is the RELATIONSHIP - data inside shape)
5. RESULT: Map showing 30-40 colored regions per state  
(User sees clear visual = temperature mapping)

#### CRITICAL PRINCIPLE:

- Every colored region = has a boundary
- Every boundary = has a temperature



- Every temperature = has a color shade
  - The visual and the geography ARE THE SAME THING
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## THE DESIGN OUTCOME

### What Your Supervisor Sees:

NOT THIS: "A heatmap with random colored patches"

BUT THIS: "A geospatial analysis showing temperature distribution across rice cultivation clusters in Nigeria, where each defined geographic region is colored according to its measured temperature value"

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## SUMMARY: THE EXACT DESIGN TRANSFORMATION

### FROM:

- Random colored patches with no geographic anchor
- User confusion: "What area does this color represent?"
- Looks unscientific

### TO:

- Colored regions with clear polygon boundaries
- User clarity: "Orange region = that cluster = 28°C"
- Looks professionally researched

### THE MECHANISM:

- Boundary (geography) + Color (temperature) = Clear relationship
- Each cluster is a polygon that gets filled with a shade
- Shades range continuously from light yellow to dark red
- UI slider is clean, backend rendering is precise

**THAT'S IT. That's the entire design concept.**